

Unit 3, Lesson 4: Writing Small Numbers in Scientific Notation

Benchmark

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Learning Targets

- ☐ I can express small numbers in scientific notation.
- ☐ I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.

3.4.1 Warm-Up: Math Mindset

In the story of *The Tortoise and the Hare*, the hare had a fixed mindset while the tortoise had a growth mindset.

1. “Failure is an almost essential step to success.” Describe a time in which you experienced failure before reaching success.
2. What can you do to overcome the thoughts of, “I’m not good at _____.” Or “I can’t _____.”?

3.4.2 Guided Instruction: Writing Small Numbers in Scientific Notation

Powers of ten can also be used to express small numbers in scientific notation.

1. A single red blood cell is about 0.000007 meters in diameter.
 - a. In what place is the 7 in the value, 0.000007?
 - b. Which power of 10 relates to this place value?
 - c. Express 0.000007 in scientific notation.

2. Consider the value 0.0000672.
- What value **between 1 and 10** could be multiplied by a **power of 10** to result in 0.0000672?
 - Which digit is in the **ones place** of your answer to Part A?
 - In which **place value** is that digit in 0.0000672?
 - Which **power of 10** relates to this place value?
 - Express 0.0000672 in **scientific notation**.
3. Consider the value 3.4×10^{-9} .
- Complete the statements.
 10^{-9} relates to the _____ place. When rewriting 3.4×10^{-9} in standard form, the ____ will be in the billionths place, followed by the _____. The places before the 3 will each hold the digit, _____.
 - Express this value in standard form.

3.4.3 Guided Practice: Writing Small Numbers in Scientific Notation

1. The density of oxygen is 1332 millionths of a gram per cubic centimeter.
 - a. Express this value as a decimal in standard form.

 - b. Express this value in scientific notation.

2. The Mycoplasmas bacteria is about 2.5×10^{-4} millimeter (mm) in size.
Express this value in standard form.

3.4.4 Collaboration: Comparing Small Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

a. Complete the table.

Scientific Notation	Standard Form
7.2×10^{-4}	
	0.00000018

b. Complete the statement to compare the factors being multiplied by powers of 10 in each number that is expressed in scientific notation.

7.2 is _____ times

- ☐ smaller
☐ larger

than _____.

c. Complete the statement to compare the powers of 10 in each number that is expressed in scientific notation.

10^{-4} is _____ times

- ☐ smaller
☐ larger

than $10^{\boxed{}}$.

d. Complete the statement.

The value 7.2×10^{-4} is _____ times

- ☐ smaller
☐ larger

than 0.00000018.

3.4.5 Your Turn!

1. Express each number in scientific notation.

a. The diameter of a grain of sand is 0.0024 inch.

b. The mass of a dust particle is about 753 trillionths of a kilogram (kg).

2. Express each number in standard form.

a. 3×10^{-8}

b. 1.2379×10^{-1}

3.4.6 Wrap-Up: Ticket Out the Door

1. Match each value on the left with its expression in scientific notation on the right by drawing lines to connect the equivalent expressions.

988 ten thousandths

$$9.88 \cdot 10^{-4}$$

0.988

$$9.88 \cdot 10^{-5}$$

0.00988

$$9.88 \cdot 10^{-2}$$

0.0000988

$$9.88 \cdot 10^{-3}$$

988 millionths

$$9.88 \cdot 10^{-1}$$